

Plastic pollution in mangrove ecosystems: impacts, mitigation actions, and policy responses

260 sources reviewed · 1993–2026 · 7 thematic areas · 6 geographic regions

- Summary of evidence
- Thematic findings
- Evidence landscape
- Knowledge gaps
- Source table

This scoping review synthesised 260 sources spanning 33 years (1993–2026) across peer-reviewed literature, grey literature, policy documents, and intergovernmental reports. The evidence base has grown rapidly: nearly half (48%) of all sources were published from 2023 onward, reflecting the emergence of mangrove plastic pollution as a distinct research field.

260

Sources reviewed

across 7 thematic areas

173

Peer-reviewed articles

67% of evidence base

86

High-quality sources

scored 90–100%

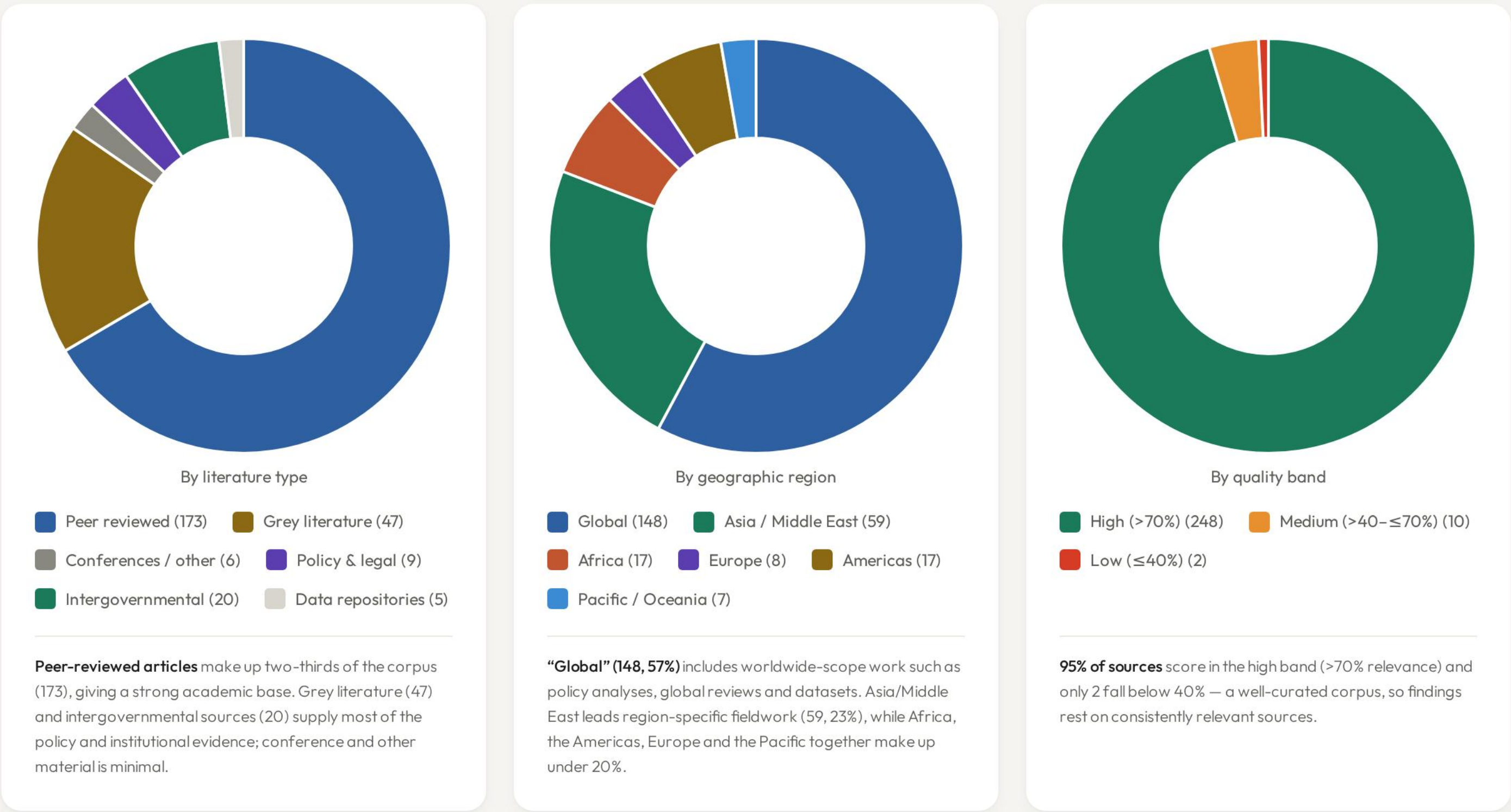
6

Regions covered

Asia/ME most studied

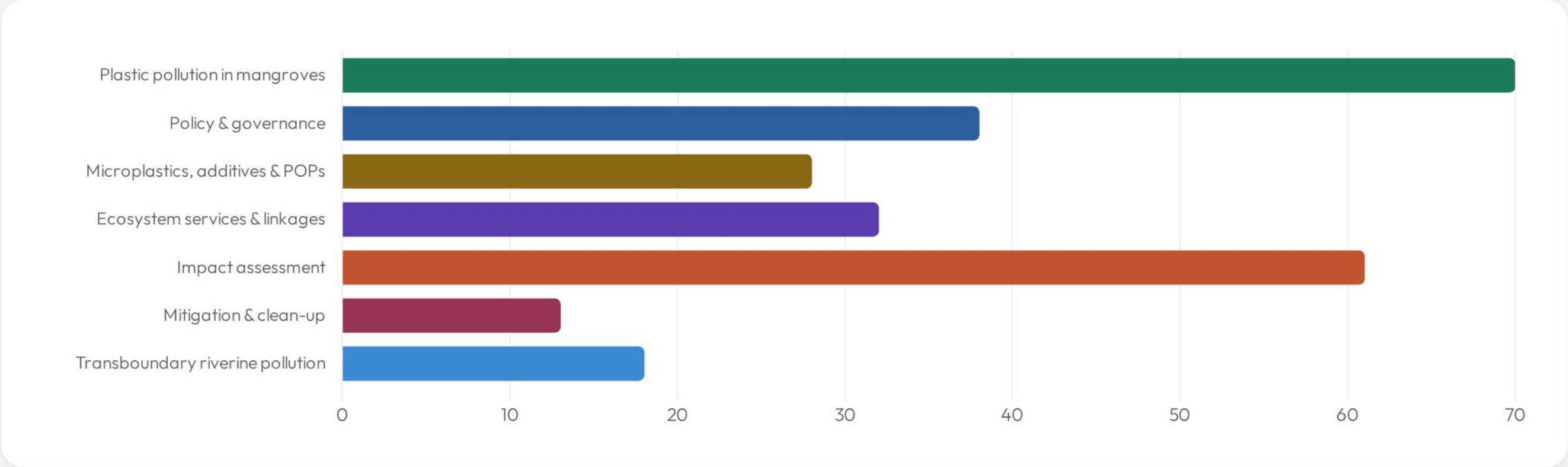
Evidence at a glance

Three key breakdowns of the 260-source corpus: literature type, geographic spread, and quality banding.



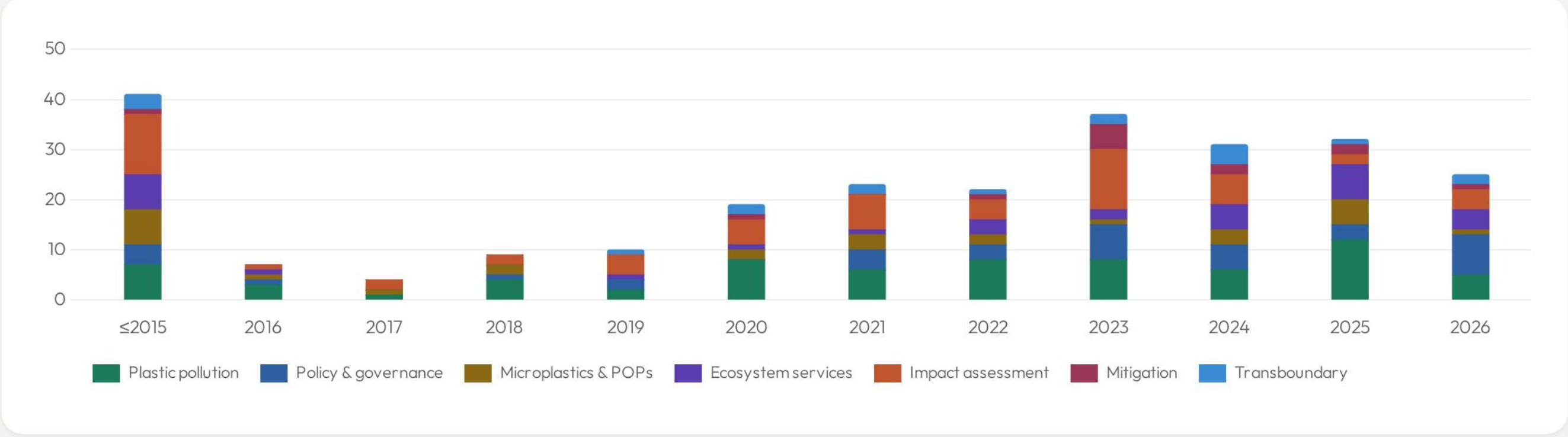
Evidence by thematic area

Volume of evidence varies significantly across research themes. Plastic pollution state assessments have the strongest evidence base; mitigation practices and transboundary dynamics have the least.



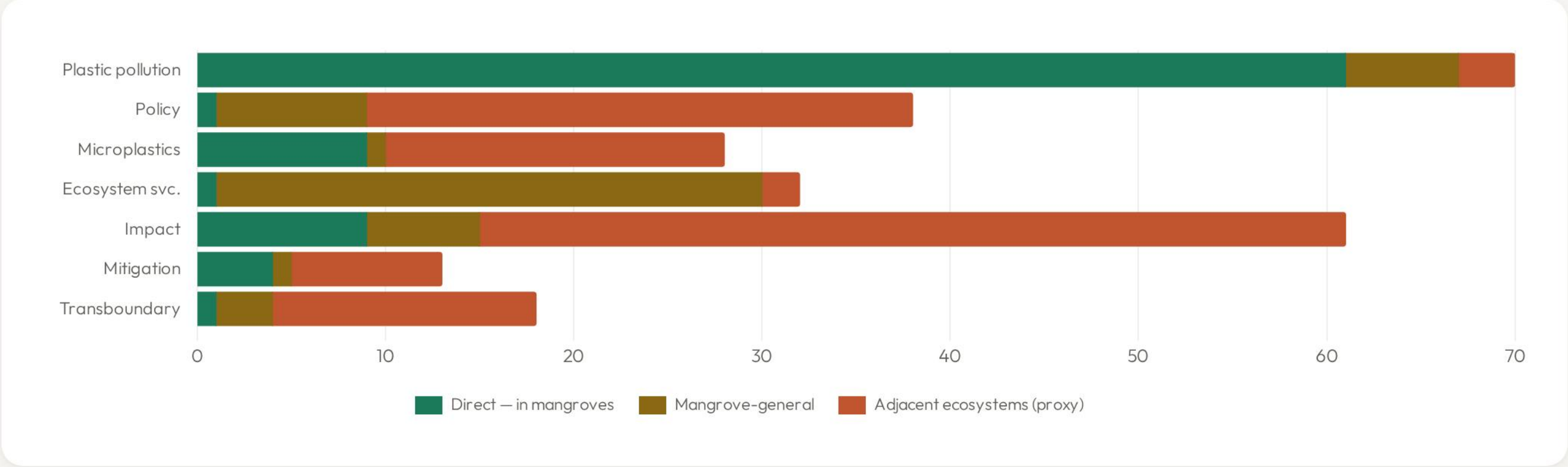
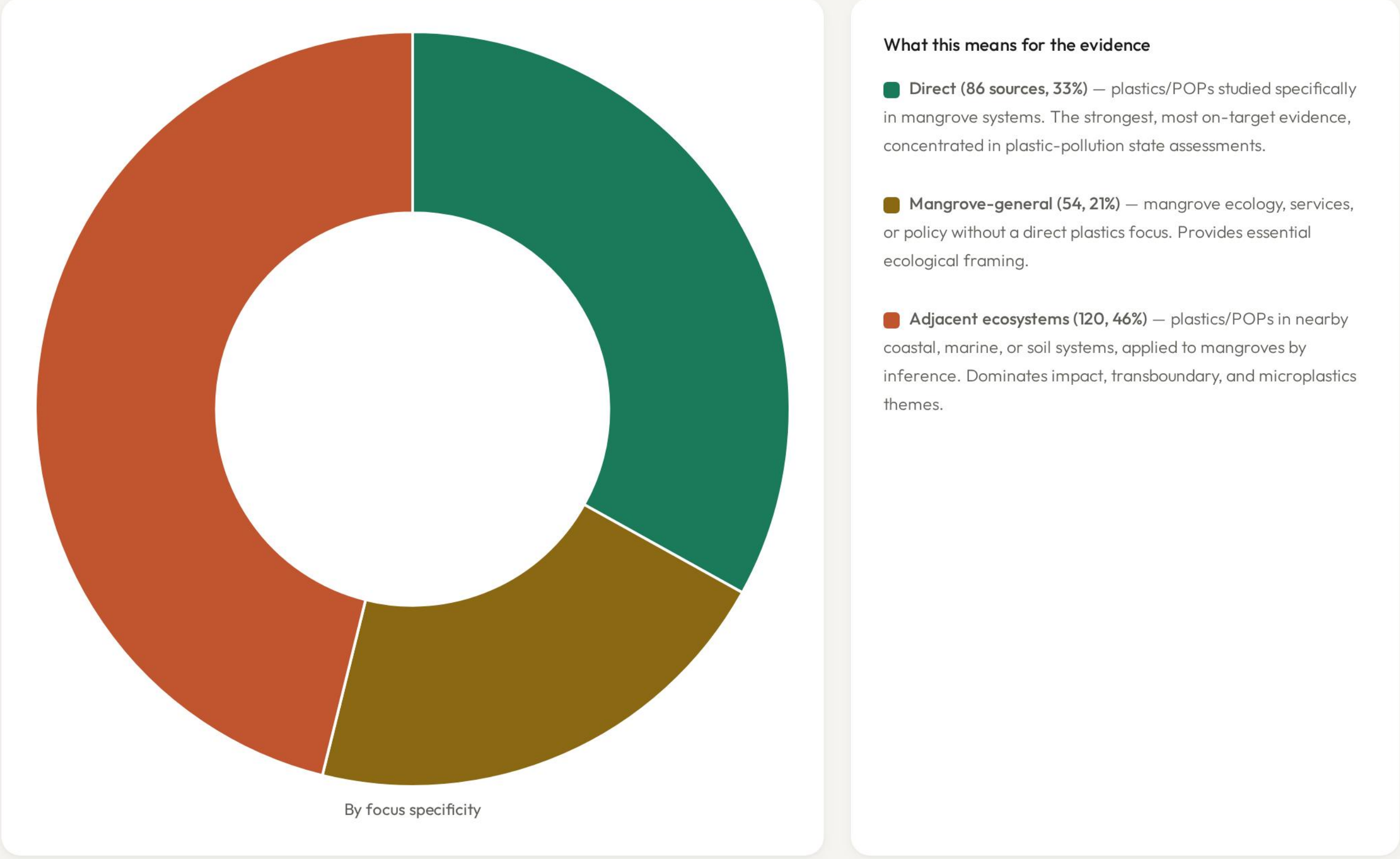
Growth of the evidence base

Research output has accelerated sharply since 2023, driven by the Global Plastics Treaty negotiations and growing recognition of mangroves as pollutant sinks. Mitigation research is the most recent theme — 77% published since 2023. Breaking the yearly output down by theme shows what drove that growth: impact assessment surged around 2023, while policy and ecosystem-services research accelerated most in 2024–2026. Each bar’s total height is the year’s overall output.



How directly does the evidence target mangroves?

A core methodological question for this review: only a third of sources study plastics or POPs *within* mangroves directly. Nearly half draw on adjacent coastal and marine ecosystems as proxy evidence — important for context, but a step removed from mangrove-specific conditions.



Direct evidence is heavily concentrated in plastic-pollution state assessments (61 of 86). Impact, transboundary, microplastics, and policy themes rely predominantly on adjacent-ecosystem proxy evidence.

Is the field moving toward direct mangrove study?

Focus specificity over time. Direct in-mangrove studies have grown in volume — from 10 before 2016 to 41 across 2023–26 — but their share of output has held near a third, because adjacent-ecosystem proxy work has grown just as fast. The corpus is expanding without becoming proportionally more mangrove-specific.

